How to Read Mobile Laser Scanner Accuracy Specifications

Nov 12, 2026 · @Mr Daryl

1. The short version

A scanner datasheet tells you what the equipment can do on its best day. It does not tell you what a survey achieved. Five rules cover most of what you need to read one correctly.

1. **Find the position accuracy, not the scanner accuracy.** The headline "5 mm" is the laser alone. Where points land on the ground depends on the position figure, which is 10 to 50 mm.
2. **Read the footnotes.** Every number assumes clear sky, post-processing and often extra hardware. The footnote tells you which.
3. **Check which statistic it is.** Manufacturers publish one sigma or RMS. FDOT sets limits at 95%. The 95% figure is about twice the published one.
4. **Convert before you compare.** Datasheets are in millimeters or meters. FDOT is in feet. ASPRS is in centimeters. A number means nothing until it is in the same unit and the same statistic as the limit.
5. **Look for the outage figure.** It shows what happens under bridges and trees. It is often ten times worse than the headline.

One line to remember: FDOT Type A allows 0.06 ft at 95%, which is 18 mm, or about the width of a penny. Type B allows 0.10 ft, which is 30 mm, or about 1¼ inches.

2. Units

Every value that appears on these datasheets is listed here in metric and imperial, so you never have to convert in your head. One millimeter is 0.00328 ft; one hundredth of a foot is 3.05 mm, or about ⅛ inch.

Millimeters	Centimeters	Meters	Feet	Inches	About the size of
3	0.3	0.003	0.010	0.12	Two stacked pennies
5	0.5	0.005	0.016	0.20	3/16 inch
10	1.0	0.010	0.033	0.39	3/8 inch

Millimeters	Centimeters	Meters	Feet	Inches	About the size of
11	1.1	0.011	0.036	0.43	7/16 inch
14	1.4	0.014	0.046	0.55	9/16 inch
16	1.6	0.016	0.052	0.63	5/8 inch
20	2.0	0.020	0.066	0.79	Width of a penny
50	5.0	0.050	0.164	1.97	2 inches
70	7.0	0.070	0.230	2.76	2¾ inches
100	10.0	0.100	0.328	3.94	4 inches

The limits you are comparing against:

Limit	Feet	Millimeters	Inches
FDOT Type A, at 95%	0.06	18.3	0.72
FDOT Type B, at 95%	0.10	30.5	1.20
Type A as vertical RMSE	0.031	9.3	0.37
Type A as horizontal RMSE	0.035	10.6	0.42
Type B as vertical RMSE	0.051	15.6	0.61
Type B as horizontal RMSE	0.058	17.6	0.69

Three unit traps:

- **Meters written small.** Trimble prints "0.020" and "0.01" in meters. That is 20 mm and 10 mm, not 0.02 mm.
- **Decimal feet are not inches.** 0.06 ft is 0.72 inch, not 0.6 inch or 6 inches.
- **Two kinds of foot.** The US survey foot and the international foot differ by 2 parts per million. That is nothing on an accuracy value, but it is 2 ft on a state plane coordinate of 1,000,000 ft. Check which foot the project uses before comparing coordinates.

3. The four kinds of accuracy on a datasheet

A datasheet lists several different accuracies, and they answer different questions. Mixing them up is the most common reading error.

Datasheet line	What it measures	Typical value	What it tells you
Scanner accuracy	How close the laser's distance reading is to the true distance	5 mm (0.016 ft)	The laser is not the limiting factor
Scanner precision	How closely repeated laser readings agree with each other	3 mm (0.010 ft)	The noise, or "fuzz", on a flat surface
Position accuracy (absolute)	How close the vehicle's computed position is to its true position on the earth	10 to 50 mm (0.033 to 0.164 ft)	Where the whole point cloud sits. This is the one that decides compliance
Roll, pitch and heading accuracy	How well the system knows which way it is tilted and pointed	0.0025° to 0.015°	An error that grows with distance from the vehicle

Two more terms you will meet:

- **Accuracy versus precision.** Accuracy is closeness to the truth. Precision is repeatability. A scanner can be precise and wrong: every pass agrees, and every pass is 20 mm high.
- **Absolute versus relative.** Absolute accuracy is position on the earth. Relative accuracy is how well nearby points agree with each other. An older RIEGL system listed 20 to 50 mm absolute and 10 mm relative. So a bridge clearance or a lane width, which are relative measurements, can be good even when the coordinates are shifted.

Common sense test: if a salesperson says "5 mm accuracy", ask "of the laser, or of the points on the ground?"

4. The manufacturers' numbers side by side

These are the published figures for the four systems, each shown in millimeters and feet. All are best-case values: clear sky, post-processed.

Position accuracy, clear sky

System	Horizontal	Vertical
Trimble MX90	Under 10 mm (0.033 ft)	10 mm (0.033 ft)
Leica Pegasus TRK Neo / Evo	11 mm (0.036 ft)	11 mm (0.036 ft)
RIEGL VMX-2HA	20 to 50 mm (0.066 to 0.164 ft)	20 to 50 mm (0.066 to 0.164 ft)
Trimble MX9	20 mm (0.066 ft)	50 mm (0.164 ft)

Position accuracy after a 60-second GNSS outage

System	Horizontal	Vertical
Trimble MX90	100 mm (0.328 ft)	70 mm (0.230 ft)
Leica Pegasus TRK Neo / Evo	14 mm (0.046 ft)	16 mm (0.052 ft)
RIEGL VMX-2HA	Not published	Not published
Trimble MX9	100 mm (0.328 ft)	70 mm (0.230 ft)

Scanner and orientation

System	Scanner accuracy / precision	Roll and pitch	Heading
Trimble MX90	5 mm / 3 mm (0.016 / 0.010 ft)	0.0025°	0.015°
Leica Pegasus TRK Neo / Evo	Not captured	Not captured	Not captured
RIEGL VMX-2HA	5 mm / 3 mm (0.016 / 0.010 ft)	0.005°	0.015°
Trimble MX9	5 mm / 3 mm (0.016 / 0.010 ft)	0.005°	0.015°

How to read these tables:

- The three scanners with published laser figures are identical at 5 mm. They differ only in positioning.
- The MX9′s vertical figure is five times its sister MX90′s. Vertical is what pavement design depends on.
- The MX90 goes from 10 mm to 100 mm horizontal in one minute without satellites. The Leica goes from 11 mm to 14 mm. On a route with bridges and canopy, the outage line matters more than the headline.
- The RIEGL position range and orientation values come from RIEGL datasheets for sister systems with the same scanner and class of positioning unit. Confirm against the VMX-2HA datasheet itself.
- A real-world reference point: Monroe County′s 2019 mobile lidar tested at 0.09 ft (27 mm) RMSE on roads, using 108 checkpoints. Field results sit well above datasheet figures unless the data is adjusted to tight control.

5. The fine print

Each published number holds only under the conditions in its footnote. Remove a condition and the number no longer applies.

Footnote wording	Plain meaning	What happens without it
"No GNSS outage"	Open sky for the whole drive	Use the outage figure instead
"Post-processed using base station data"	The path was recomputed in the office against a fixed reference receiver	Real-time results are worse
"With DMI option"	A wheel-mounted distance sensor was fitted	Drift is larger, mainly in outages
"With GAMS option, 2 m baseline"	A second GNSS antenna was fitted 2 m (6.6 ft) from the first	Heading accuracy is worse
"One sigma"	About 68% of results fall within the number	See section 6
"At 30 m range under test conditions"	The laser figure was measured at 30 m (98 ft) on a test target	Longer ranges and dark or wet surfaces are noisier

Footnote wording	Plain meaning	What happens without it
"Measured in a controlled test area"	The maker's own site and procedures	Your project is not that site
"Typical" or "typ."	An expected value, not a guarantee	Results can be worse
"Target reflectivity above 80%"	Maximum range assumes a bright white target	Asphalt is around 10 to 20%; range is much shorter

Which footnotes go with which maker:

- **RIEGL** states its position and orientation values are one sigma, no GNSS outage, with the distance sensor, post-processed. Its scanner figure is one sigma at 30 m.
- **Trimble** states the MX90 position values are post-processed, with the distance sensor, in a controlled test area, and heading with the second antenna.
- **Leica** states "post-processing" and "no GNSS outage" in the table. I did not capture its footnote on the statistic used.

Range also trades against speed. The MX90 reaches 475 m (1,558 ft) at its slowest pulse rate and 235 m (771 ft) at higher rates, both on bright targets. For road work this hardly matters: pavement is within about 20 m (66 ft).

6. One sigma, RMSE and 95%

A published accuracy of 10 mm does not mean every point is within 10 mm. It means about two thirds are. FDOT asks where 95% fall, and that number is roughly double.

Term	Plain meaning	Share of results inside it
One sigma	The typical size of the error	About 68% (about 2 in 3)
RMSE	The typical size of the error, measured against truth. Equal to one sigma when there is no bias	About 68%
95% confidence	The size that 19 results out of 20 stay inside	95%

To go from a published one-sigma or RMSE value to the 95% value:

- Vertical: multiply by 1.96.
- Horizontal: multiply by 1.7308 if the figure is a combined (radial) value, or by 2.4477 if it is per axis (east and north each). Datasheets rarely say which, so ask.

Published vertical figures converted to FDOT terms

System	Published vertical	At 95%	In feet at 95%	Type A (0.06 ft)	Type B (0.10 ft)
Trimble MX90	10 mm	19.6 mm	0.064 ft	Just over	Within
Leica Pegasus TRK	11 mm	21.6 mm	0.071 ft	Over	Within
RIEGL VMX-2HA (best case)	20 mm	39.2 mm	0.129 ft	Over	Over
Trimble MX9	50 mm	98.0 mm	0.322 ft	Over	Over

Published horizontal figures converted to FDOT terms

System	Published horizontal	At 95% if radial	At 95% if per axis	Type A (0.06 ft)
Trimble MX90	10 mm	17.3 mm (0.057 ft)	24.5 mm (0.080 ft)	Within only if radial
Leica Pegasus TRK	11 mm	19.0 mm (0.062 ft)	26.9 mm (0.088 ft)	Just over, or over
RIEGL VMX-2HA (best case)	20 mm	34.6 mm (0.114 ft)	49.0 mm (0.161 ft)	Over
Trimble MX9	20 mm	34.6 mm (0.114 ft)	49.0 mm (0.161 ft)	Over

What this shows:

- Even the best published figure, 10 mm, becomes 0.064 ft at 95%. That is over the Type A limit of 0.06 ft before any other error is added.
- This is why no unit meets Type A without ground control, and why a "1 cm system" is not a "0.03 ft at 95%" system.

- These conversions assume the published figure is one sigma or RMSE with no bias. RIEGL says one sigma. For Trimble and Leica I did not capture the statistic, so treat their rows as a reasonable reading, not a certainty.

7. Angles into distance

An angle error is small near the vehicle and large far from it. To turn degrees into distance, multiply the range by the tangent of the angle.

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\text{error} = \text{range} \times \tan(\text{angle})
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Angle accuracy	At 20 m (66 ft)	At 50 m (164 ft)	At 100 m (328 ft)
0.0025° (MX90 roll and pitch)	0.9 mm (0.003 ft)	2.2 mm (0.007 ft)	4.4 mm (0.014 ft)
0.005° (MX9, RIEGL roll and pitch)	1.7 mm (0.006 ft)	4.4 mm (0.014 ft)	8.7 mm (0.029 ft)
0.015° (heading, all three)	5.2 mm (0.017 ft)	13.1 mm (0.043 ft)	26.2 mm (0.086 ft)

Worked: heading 0.015° at 50 m is 50,000 mm × tan(0.015°) = 50,000 × 0.000262 = 13.1 mm.

What this means in practice:

- **Pavement is safe.** The road surface is within about 20 m (66 ft), where angle errors are 1 to 5 mm.
- **Far features are not.** A sign or building corner 100 m (328 ft) away carries 26 mm (0.086 ft) from heading alone. That exceeds Type A before anything else is counted.
- **Roll and pitch mostly move points up and down.** Heading mostly moves them sideways.
- **Set an extraction limit.** For Type A, take measurements only within a stated distance of the vehicle path, and say what that distance is in the survey report.

8. Putting it together: an error budget

The accuracy of a point on the ground combines all four datasheet lines. Independent errors combine as the square root of the sum of their squares, so the largest one dominates.

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\text{total} = \sqrt{\text{position}^{2} + \text{scanner}^{2} +
\text{roll/pitch}^{2} + \text{heading}^{2}}
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A point 20 m (66 ft) from the vehicle, straight off the datasheet

Error source	Trimble MX90	RIEGL VMX-2HA (best case)
Position	10 mm	20 mm
Scanner	5 mm	5 mm
Roll and pitch at 20 m	0.9 mm	1.7 mm
Heading at 20 m	5.2 mm	5.2 mm
Total, one sigma	12.4 mm (0.041 ft)	21.3 mm (0.070 ft)
Total at 95%	24.3 mm (0.080 ft)	41.8 mm (0.137 ft)
FDOT Type A, 0.06 ft	Over	Over
FDOT Type B, 0.10 ft	Within	Over

Worked for the MX90: $\sqrt{(10^2 + 5^2 + 0.9^2 + 5.2^2)} = \sqrt{153.2} = 12.4$ mm. Then 12.4 × 1.96 = 24.3 mm, and 24.3 ÷ 304.8 = 0.080 ft.

The same point after adjustment to ground control

Control replaces the position term with what is left after adjustment. Assume 5 mm remains.

Error source	Trimble MX90	RIEGL VMX-2HA
Position after control	5 mm	5 mm
Scanner, roll and pitch, heading	Unchanged	Unchanged
Total, one sigma	8.8 mm (0.029 ft)	9.0 mm (0.029 ft)
Total at 95%	17.3 mm (0.057 ft)	17.6 mm (0.058 ft)
FDOT Type A, 0.06 ft	Within	Within

What the budget shows:

- Off the datasheet, the two systems differ by almost a factor of two. After control, they are the same to within 0.2 mm.
- After control, the laser and the heading are the largest terms. That is the floor, and it sits just inside Type A.
- The margin is thin: 0.057 ft against 0.06 ft. Good checkpoints and short extraction ranges are what keep it there.
- This is a simplified estimate. It treats all errors as acting in one direction and assumes 5 mm after control. A real result comes only from the checkpoint test.

9. Reading a survey report

A survey report states what was achieved, and the same result can be written three ways. Know which form you are looking at before comparing it to a limit.

One result, three forms (from the sample 32-checkpoint test in the accuracy workbook)

Form	Vertical	Horizontal	Compare it to
RMSE in metric (ASPRS style)	0.71 cm (7.1 mm)	0.81 cm (8.1 mm)	The ASPRS accuracy class in cm
RMSE in feet	0.023 ft	0.027 ft	The RMSE targets in section 2
95% confidence in feet (FDOT style)	0.046 ft	0.046 ft	The FDOT limit: 0.06 or 0.10 ft

All three rows describe the same data, and all three pass Type A.

Common mistakes

Mistake	Why it is wrong
Comparing an RMSE to the FDOT limit	The FDOT limit is at 95%. An RMSE of 0.05 ft looks like a pass against 0.06 ft but is 0.098 ft at 95%
Reading 0.71 cm as 0.71 ft	Off by a factor of 30. Check the unit on every number

Mistake	Why it is wrong
Treating the datasheet figure as the result	The datasheet describes equipment. Only checkpoints describe a survey
Accepting control residuals as accuracy	Points used to adjust the data always fit well
Ignoring the checkpoint count	ASPRS needs 30; FDOT needs 20. Fewer means a qualified result
Ignoring the mean error	A mean far from zero means a bias, and the 95% conversion no longer holds

Questions to ask a vendor or surveyor

1. Is this number for the laser or for the points on the ground?
2. Is it one sigma, RMSE or 95%?
3. Is the horizontal figure radial or per axis?
4. Which options were fitted: wheel sensor, second antenna, which inertial unit?
5. What is the figure after a GNSS outage, and how long were the outages on this project?
6. How many independent checkpoints were used, and how accurately were they surveyed?
7. What is the tested result in feet at 95%, horizontal and vertical?
8. Within what distance of the vehicle path does that result apply?

Sources:

- Trimble MX90 spec sheet
- Trimble MX9 datasheet
- Leica Pegasus TRK product page; accuracy figures from the Pegasus TRK Neo & Evo data sheet
- RIEGL VMX-2HA product page, VMX-1HA datasheet and VMX-RAIL brochure for position, orientation and footnotes
- Monroe County mobile lidar accuracy assessment, 2019
- FDOT Surveying and Mapping Handbook for the Type A and B limits; multipliers 1.96 and 1.7308 are the NSSDA values

Datasheets are revised often. Check each figure against the current datasheet before quoting it in a contract or report.